

Assignment No. 5

Subject: Database Engineering Lab (7CS271)

Class: SY CSE, AY 2025–26, Sem-II

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Batch - S5

1. BASIC DATA RETRIEVAL

- a. Display all records from a table

Query

```
SELECT * FROM Person;
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
6	Ankit	Verma	11	Street	Pune	411001	01/01/1991	a6@gmail.com
7	Neha	Joshi	22	Road	Mumbai	400001	02/02/1992	a7@gmail.com
8	Rahul	Kapoor	33	Lane	Delhi	110001	03/03/1988	a8@gmail.com
9	Priya	Nair	44	Circle	Chennai	600001	04/04/1993	a9@gmail.com
10	Vikram	Reddy	55	Avenue	Hyderabad	500001	05/05/1987	a10@gmail.com
3	Rohan	Mehta	22	Link Road	Mumbai	400050	01/20/1990	a3@gmail.com
5	Karan	Singh	10	Civil Lines	Delhi	110001	11/25/1992	a5@gmail.com
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com
2	Diya	Patel	45	FC Road	Pune	411004	08/15/2003	a2@gmail.com
4	Sneha	Iyer	18	Brigade Road	Bangalore	560001	03/12/1985	a4@gmail.com

b. Display specific columns

Query

```
SELECT first_name, city FROM Person;
```

Result

FIRST_NAME	CITY
Ankit	Pune
Neha	Mumbai
Rahul	Delhi
Priya	Chennai
Vikram	Hyderabad
Rohan	Mumbai
Karan	Delhi
Aarav	Pune
Diya	Pune
Sneha	Bangalore

c. Remove duplicate values using DISTINCT

Query

```
SELECT DISTINCT city FROM Person;
```

Result

CITY
Delhi
Mumbai
Bangalore
Hyderabad
Pune
Chennai

d. Rename output columns using aliases

Query

```
SELECT first_name AS "First Name", city AS "City"  
  
FROM Person;
```

Result

First Name	City
Ankit	Pune
Neha	Mumbai
Rahul	Delhi
Priya	Chennai
Vikram	Hyderabad
Rohan	Mumbai
Karan	Delhi
Aarav	Pune
Diya	Pune
Sneha	Bangalore

2. FILTERING RECORDS

a. WHERE with comparison operators

Query

```
SELECT * FROM Student
```

```
WHERE cgpa > 8;
```

```
SELECT * FROM Course
```

```
WHERE credits >= 4;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2

COURSE_ID	TITLE	COURSE_CODE	CREDITS
501	DBMS	CS301	4
502	OS	CS302	4
504	CN	CS303	4

b. Logical operators (AND, OR, NOT)

Query

```
SELECT * FROM Person
```

```
WHERE city = 'Pune' AND first_name = 'Aarav';
```

```
SELECT * FROM Person
```

```
WHERE city = 'Pune' OR city = 'Mumbai';
```

```
SELECT * FROM Student
```

```
WHERE NOT cgpa < 7;
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
6	Ankit	Verma	11	Street	Pune	411001	01/01/1991	a6@gmail.com
7	Neha	Joshi	22	Road	Mumbai	400001	02/02/1992	a7@gmail.com
3	Rohan	Mehta	22	Link Road	Mumbai	400050	01/20/1990	a3@gmail.com
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com
2	Diya	Patel	45	FC Road	Pune	411004	08/15/2003	a2@gmail.com

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
103	3	07/01/2021	-	6	7.8
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2
105	5	07/01/2020	-	8	7.5

c. BETWEEN

Query

```
SELECT * FROM Student  
  
WHERE cgpa BETWEEN 7 AND 9;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
103	3	07/01/2021	-	6	7.8
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2
105	5	07/01/2020	-	8	7.5

d. IN

Query

```
SELECT * FROM Person  
  
WHERE city IN ('Pune','Mumbai','Delhi');
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
6	Ankit	Verma	11	Street	Pune	411001	01/01/1991	a6@gmail.com
7	Neha	Joshi	22	Road	Mumbai	400001	02/02/1992	a7@gmail.com
8	Rahul	Kapoor	33	Lane	Delhi	110001	03/03/1988	a8@gmail.com
3	Rohan	Mehta	22	Link Road	Mumbai	400050	01/20/1990	a3@gmail.com
5	Karan	Singh	10	Civil Lines	Delhi	110001	11/25/1992	a5@gmail.com
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com
2	Diya	Patel	45	FC Road	Pune	411004	08/15/2003	a2@gmail.com

e. LIKE

Query

```
SELECT * FROM Person
```

```
WHERE first_name LIKE 'A%';
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
6	Ankit	Verma	11	Street	Pune	411001	01/01/1991	a6@gmail.com
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com

f. NULL handling

Query

```
SELECT * FROM Student
```

```
WHERE graduation_date IS NULL;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
103	3	07/01/2021	-	6	7.8
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9
104	4	07/01/2021	-	6	8.2
105	5	07/01/2020	-	8	7.5

3. SORTING AND ORDERING

a. ORDER BY (Ascending)

Query

```
SELECT * FROM Student
```

```
ORDER BY cgpa ASC;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
105	5	07/01/2020	-	8	7.5
103	3	07/01/2021	-	6	7.8
104	4	07/01/2021	-	6	8.2
101	1	07/01/2022	-	4	8.5
102	2	07/01/2022	-	4	9

b. ORDER BY (Descending)

Query

```
SELECT * FROM Student
```

```
ORDER BY cgpa DESC;
```

Result

STUDENT_ID	PERSON_ID	ENROLLMENT_DATE	GRADUATION_DATE	SEMESTER	CGPA
102	2	07/01/2022	-	4	9
101	1	07/01/2022	-	4	8.5
104	4	07/01/2021	-	6	8.2
103	3	07/01/2021	-	6	7.8
105	5	07/01/2020	-	8	7.5

c. Sorting by multiple columns

Query

```
SELECT * FROM Person
```

```
ORDER BY city ASC, first_name ASC;
```

Result

PERSON_ID	FIRST_NAME	LAST_NAME	STREET_NO	STREET_NAME	CITY	ZIP_CODE	DATE_OF_BIRTH	EMAIL
4	Sneha	Iyer	18	Brigade Road	Bangalore	560001	03/12/1985	a4@gmail.com
9	Priya	Nair	44	Circle	Chennai	600001	04/04/1993	a9@gmail.com
5	Karan	Singh	10	Civil Lines	Delhi	110001	11/25/1992	a5@gmail.com
8	Rahul	Kapoor	33	Lane	Delhi	110001	03/03/1988	a8@gmail.com
10	Vikram	Reddy	55	Avenue	Hyderabad	500001	05/05/1987	a10@gmail.com
7	Neha	Joshi	22	Road	Mumbai	400001	02/02/1992	a7@gmail.com
3	Rohan	Mehta	22	Link Road	Mumbai	400050	01/20/1990	a3@gmail.com
1	Aarav	Sharma	12	MG Road	Pune	411001	05/10/2004	a1@gmail.com
6	Ankit	Verma	11	Street	Pune	411001	01/01/1991	a6@gmail.com
2	Diya	Patel	45	FC Road	Pune	411004	08/15/2003	a2@gmail.com

4. COMPUTED COLUMNS & EXPRESSIONS

a. Arithmetic operation

Query

```
SELECT student_id, cgpa, cgpa + 0.5 AS "Improved CGPA"  
FROM Student;
```

Result

STUDENT_ID	CGPA	Improved CGPA
103	7.8	8.3
101	8.5	9
102	9	9.5
104	8.2	8.7
105	7.5	8

b. Concatenate text fields

Query

```
SELECT first_name || ' ' || last_name AS "Full Name"  
FROM Person;
```

Result

Full Name
Ankit Verma
Neha Joshi
Rahul Kapoor
Priya Nair
Vikram Reddy
Rohan Mehta
Karan Singh
Aarav Sharma
Diya Patel
Sneha Iyer

c. Derived values

Query

```
SELECT course_id, credits, credits * 25 AS "Total Hours"  
  
FROM Course;
```

Result

COURSE_ID	CREDITS	Total Hours
501	4	100
502	4	100
504	4	100
503	3	75
505	3	75

5. Extra for Understanding Joins (By me)

a. JOIN (Student + Person)

Query

```
SELECT s.student_id, p.first_name, p.city, s.cgpa  
FROM Student s  
JOIN Person p ON s.person_id = p.person_id;
```

Result

STUDENT_ID	FIRST_NAME	CITY	CGPA
103	Rohan	Mumbai	7.8
101	Aarav	Pune	8.5
102	Diya	Pune	9
104	Sneha	Bangalore	8.2
105	Karan	Delhi	7.5

b. JOIN (Faculty + Course)

Query

```
SELECT f.faculty_id, c.title  
FROM Faculty f  
JOIN Teaches t ON f.faculty_id = t.faculty_id  
JOIN Course c ON t.course_id = c.course_id;
```

Result

FACULTY_ID	TITLE
201	DBMS
202	OS
203	DSA
204	CN
205	SE